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MILITARY STANDARD

INTERFACE STANDARD FOR SHIPBOARD SYSTEMS

SECTION 204A

AMBIENT AIR CONDITIONS IN SURFACE SHIP COMPARTMENTS (METRIC)



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FSC 1990

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DEPARTMENT OF THE NAVY
NAVAL SEA SYSTEMS COMMAND

Washington, DC 20362-5101

Interface Standard for Shipboard Systems,
Ambient Air Conditions in Surface Ship Compartments (METRIC)

DOD-STD-1399(NAVY)
SECTION 204A

1. This Military standard is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Prposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

1. Purpose. This section defines the standard interface requirements for, and the constraints on, the design of shipboard systems and equipments which are imposed by the ambient air conditions in surface ship compartments.

2. Nature of the interface. Systems and equipment on board ships are exposed to the ambient air conditions of the compartments in which they are installed. These conditions are controlled to a varying degree, by ventilation, heating, and air conditioning. Systems and equipment should be designed (and at times protected) to prevent performance degradation by the ambient air conditions surrounding them during their expectant life cycle.

3. Structure of this section. The technical content first delineates the characteristics in terms of temperature, relative humidity, rate of change of air, and other pertinent factors. The constraints on systems and equipment design and installation which are necessary to achieve shipboard compatibility with these characteristics are then established.

4. Numerical quantities. The numerical quantities stated in this section are expressed in metric (S.I.) units followed by their common equivalents in parentheses. The relationship between S.I. units and common equivalents is approximated to a practical number of significant figures.

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1. GENERAL, SCOPE, INTERFACE AND APPLICABILITY

1.1 General. Procedures established by DOD-STD-1399 are mandatory. This section and the basic standard are to be viewed as an integral single document.

1.2 Scope. This section establishes interface requirements for shipboard systems and equipment with respect to the ambient air conditions of surface ship compartments to ensure compatibility between such systems and equipment and the surrounding air environment of the compartments in which they are installed.

1.3 Interface. The basic characteristic and constraint categories concerned with this interface are shown symbolically on figure 1 (see definitions of DOD-STD-1399).

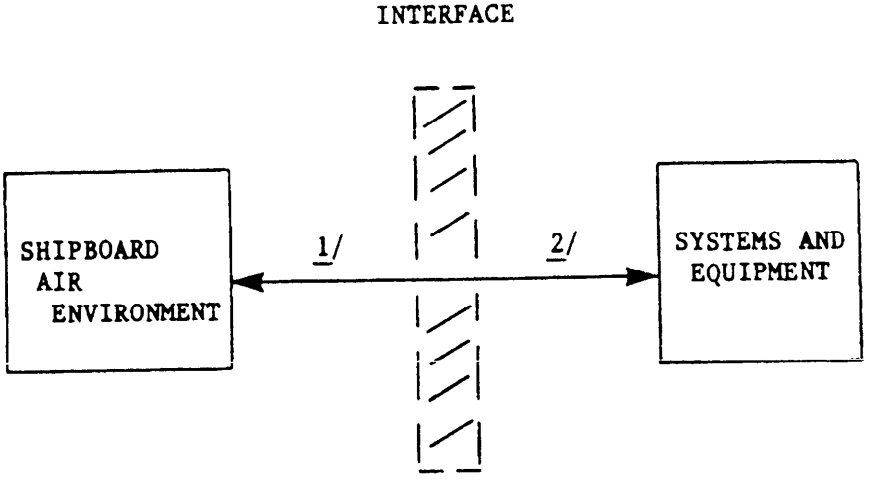


FIGURE 1. Interface.

The particular interface characteristics and constraints pertinent to this section are described in 5.2 and 5.3.

1.4 Applicability. Shipboard systems and equipment shall be capable of continuous reliable operation when exposed to the various enviornmental conditions of the air in shipboard compartments. Human engineering and shipboard habitability considerations are also of concern. This section applies to ventilated, heated, and air conditioned compartments in surface ships and to the design and installation of systems and equipment destined for such spaces. It does not apply to reactor compartments of nuclear powered ships.

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2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specification and standards of the issue listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-16400 - Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for.

STANDARDS

MILITARY

MIL-STD-454 - Standard General Requirements for Electronic Equipment.

DOD-STD-1399 - Interface Standard for Shipboard Systems. (Metric)

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

3. DEFINITIONS

3.1 Compartment ambient conditions. Compartment ambient conditions are the temperature and relative humidity prevailing in a shipboard compartment.

3.2 Air conditioning. Air conditioning is the control of dry bulb temperature and limitation of relative humidity by means of mechanically cooled, recirculated air.

3.3 Replenishment air. Replenishment air is weather air that is supplied to air conditioned compartments to maintain an acceptable level of air purity.

3.4 Ventilation. Ventilation is the movement of air from the weather into the ship or from the ship out to the weather by fans or ductwork, or both.

3.4.1 Zone heating and cooling. Zone heating and cooling is the use of a single reheater or cooling coil to serve a group of spaces each of which has approximately the same exposures, use, concurrent load variations, and delivery air temperature requirements.

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4. REQUIREMENTS

4.1 The specific interface requirements and constraints established herein are mandatory and shall be adhered to by SYSCOMs, Project Managers PMs, contractors and all others engaged in any aspect of shipboard ventilation, heating, and air conditioning system design to which these requirements and constraints apply including systems and equipment design, production, and installation (see requirements DOD-STD-1399).

5. INTERFACE CHARACTERISTICS AND CONSTRAINTS

5.1 General considerations. Naval ships are customarily designed to operate in any oceanic area and to perform their assigned mission under a wide variety of climatic conditions. The ambient conditions in shipboard compartments have a direct effect upon the operating efficiency of shipboard systems and equipment and upon the health, welfare and efficiency of ship personnel. In addition to the effects of the weather environment, ship generated temperature and moisture conditions present very important and sometimes overriding considerations. To meet the demands for the control of shipboard ambient conditions, ships are very carefully designed with respect to air conditioning, ventilating, and heating facilities and systems. The system design for each ship (or class) will vary depending on individual ship characteristics and requirements that are given in detail in Ships Specifications for the particular ship (or class). Individual ship designs tend to become dated; recent ships normally have more sophisticated systems than those in ships built in earlier years. These individual system designs are based on broad general criteria of a practical nature and tailored to the requirements of particular types of compartments or spaces. In specific cases, further requirements may be imposed to meet particular needs.

5.2 Interface characteristics. The interface characteristics of the ships compartment air environment are specified in 5.2.1 through 5.2.3. Each shipboard air conditioning, heating, and ventilating system provides air of the specified characteristics to designated compartments throughout the ship.

5.2.1 Air conditioned compartments. Compartments designated for air conditioning are, in general, those in which the comfort, welfare and efficient functioning of personnel are of concern, those in which certain critical systems and equipment, particularly electronic types, are installed, and certain compartments with specific rigid environmental controls which are required for the proper performance of the functions assigned. The specified conditions are design compartment ambient temperature and relative humidity for shipboard air conditioned areas. In addition, the injection of replenishment outside air may be specified to maintain an acceptable level of air purity when required.

5.2.1.1 Standard criteria. The standard criteria for ship compartments designated for air conditioning are as follows:

(a) Temperature. The design ambient temperature is 27 degrees Celsius (°C) 80 degrees Fahrenheit (°F) dry bulb during the cooling season, 18°C (65°F) during the heating season.

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(b) Relative humidity. The design ambient relative humidity is a maximum of 55 percent and a practical minimum of 35 percent during the cooling season. No lower control limits are provided during the heating season.
(c) Replenishment air. Replenishment air is supplied at a minimum rate of 0.15 cubic meters (5 cubic feet) per minute per man based on total design occupancy.
(d) Air distribution. Distribution and circulation of air prevents local dead spots where air is not changed.

5.2.1.2 Modified criteria. In certain cases the standard criteria are modified to meet the particular needs or operational requirements of individual compartments. Some examples of compartments to which modified criteria may apply are medical spaces, laboratories, vital shops, and missile service areas.

5.2.2 Ventilated compartments. Compartments not designated for air conditioning may be provided with ventilation, including zone heating and cooling, to maintain temperature and air circulation control within acceptable limits. The specified conditions are design compartment ambient temperature and rate of change of air. In machinery spaces, assumed temperatures are specified for calculation of boundary heat loss.

5.2.2.1 Standard criteria. The standard criteria for ventilated ship compartments are as follows:

(a) Temperature. The design ambient temperature is generally 38 to 41°C (100 to 105°F) during the cooling season and 18°C (65°F) during the heating season. Variation in these values may occur depending on the location and function of the compartment concerned (see 5.2.3).
(b) Rate of change of air. The designed rate of change of compartment air (stated in minutes for complete replenishment of one compartment volume of air), where specified, is related to estimated temperature rise and physiological needs within a particular compartment.
(c) Relative humidity. Relative humidity control limits are not specified for ventilated compartments.

5.2.2.2 Machinery space criteria. In machinery spaces containing heat-producing equipment, boundary effect on adjacent compartments is paramount. The temperature in such machinery compartments is not controlled, but assumed temperatures are specified for calculation of expected heat transmission loss (see 5.2.3).

5.2.3 Detailed design criteria. The design of the air conditioning, heating, and ventilating systems vary widely for different ships. System designs are tailored to specialized requirements of the particular compartment or to the needs of installed equipment (see 6.1). The system design details and compartment criteria are specified in the Ship Specifications for the applicable ship.

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5.3 Interface constraints. The interface characteristics of compartment ambient conditions impose certain constraints on the design of systems and equipment exposed to this environment. These constraints are specified in 5.3.1 through 5.3.4.

5.3.1 Compatibility. The design of systems and equipment affected by the compartment ambient conditions shall be compatible with the interface characteristics given in 5.2.

5.3.2 Moisture. Equipment design shall avoid pockets, wells, traps, and so forth in which moisture can collect or provisions shall be made for drainage. (For electronic equipment see MIL-STD-454, requirement 31.) External condensation on equipment and ductwork shall be controlled to the degree practical.

5.3.3 Natural cooling. Equipment design shall take maximum advantage of natural cooling utilizing the compartment ambient as the heat sink. (For electronic equipment see the thermal design requirements of MIL-E-16400.)

5.3.4 Special considerations. Equipment having special compartment ambient requirements not covered by the criteria of the Ship Specification shall be reported as specified in 6.1.

6. DOCUMENTATION REQUIREMENTS

6.1 DD Form 1423. When this standard is invoked, the data requirements of 7.1, when applicable, shall be specified by the Principal Development Activity (PDA) on DD Form 1423 (Contract Data Requirements List) attached to the contract or order.

7. DEVIATIONS

7.1 Conditions. In achieving the purpose of this section it is recognized that there must be some flexibility of application. During the early design stage of shipboard systems and equipment exposed to compartment ambient air conditions it may become apparent that significant advantages in the overall design and operation of such systems and equipment can be achieved by deviating from the standard characteristics specified herein. In such instance, the provisions of the deviations requirements of DOD-STD-1399 shall be complied with.

7.1.1 Deviation procedure. Requests for deviation shall contain the following information:

(a) System and equipment identification.
(b) Applicable space (if known).
(c) Ambient requirements: temperature and relative humidity.
(d) Amplifying information, as appropriate to justify deviations.

Requests for deviations shall be submitted to NAVSEA with copies to the Program/ Project Manager and NAVSEA 56Y1.

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8. NOTES

8.1 Subject term (key word) listing.

Air environment
Relative humidity
Temperature
Ventilation

8.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Review activities:
OS, EC, AS

User activity:
CG

Preparing activity:
Navy - SH
(Project 1990-N053)

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